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Thalavapalayam, Karur - 639 113, TAMILNADU.



ECC1376 - ASIC DESIGN

END SEMESTER REVIEW

Design and Verilog Implementation of a Byte-Enabled RAM Architecture for Efficient Data Access

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ABSTRACT

- Modern digital systems require efficient memory architectures to handle high-speed data processing with minimal power consumption.
- Byte-enabled RAM is an advanced memory design that allows selective access and modification of specific bytes within a word, rather than reading or writing the entire word.
- This improves memory efficiency, reduces unnecessary data operations, and enhances overall system performance. In this work, a byte-enabled RAM architecture is designed and implemented using Verilog HDL.
- The proposed system supports flexible data access, making it suitable for applications such as embedded systems, processors, and FPGA-based designs.
- Simulation results demonstrate improved efficiency in terms of speed and resource utilization compared to conventional RAM architectures.

INTRODUCTION

- Memory is used to store data and instructions in digital systems. Traditional RAM works with fixed word sizes.
- It accesses the entire word even if only a small part is needed This causes high power consumption and inefficiency.
- Byte-enabled RAM allows access to individual bytes. It improves speed and reduces unnecessary operations. Useful in modern processors and embedded systems.
- Verilog HDL is used for design due to flexibility. Suitable for FPGA and ASIC implementation. The goal is to improve efficiency with a simple and scalable design.

PROBLEM STATEMENT

Conventional RAM systems are inefficient because they always access the entire word even when only a small part of data is needed. This increases power consumption, wastes memory usage, and reduces system performance. Therefore, there is a need to design a memory system that allows selective byte access while maintaining high performance and low complexity.

OBJECTIVE

- The main objective of this project is to design and implement a byte-enabled RAM architecture using Verilog HDL.
- The goals include improving memory efficiency by enabling selective byte access, reducing power consumption, and enhancing data processing speed.
- Another objective is to develop a scalable and flexible design that can be easily integrated into FPGA-based systems.
- The design should also be verified through simulation and synthesis to ensure correctness and reliability in real-time applications.

LITERATURE SURVEY

S.No	Title of Paper	Author(s) / Year	Methodology / Technique Used	Key Findings	Limitations
1	Design and Implementation of SRAM Using Verilog	Kirti Bindal et al., 2024	SRAM modeled using Verilog HDL for ASIC applications	Improved memory modeling, better speed and reliability using HDL-based SRAM design	Does not focus on byte-enabled access mechanism
2	Power Efficient Data-Aware SRAM Cell for SRAM-Based FPGA Architecture	Ajay Kumar Singh, 2024	Data-aware SRAM cell with modified decoder and sensing circuits	Reduced leakage power and improved FPGA memory efficiency	Complex peripheral circuitry increases design complexity
3	Design of SRAM and DRAM Volatile Memories using 45nm Technology for FPGA Architecture	Hamsa S. et al., 2025	SRAM and DRAM implementation using CMOS 45nm technology	Reduced delay and power consumption compared to 180nm design	No byte-level selective write operation support
4	A Survey of SRAM-based In-Memory Computing Techniques and Applications	Sparsh Mittal et al., 2025	SRAM-based in-memory computing architecture survey	Reduced data movement overhead and improved computational efficiency	Primarily focused on processing-in-memory rather than RAM accessibility
5	SRAM Cell Leakage Control Techniques for Ultra Low Power Application	Pavankumar Bikki et al. 2025	Leakage reduction techniques for SRAM cells	Significant reduction in standby power consumption	Trade-off between stability and performance

EXISTING SYSTEM

- The existing RAM architecture is based on a conventional word-level memory access mechanism where the entire memory word is read or written at once.
- In most traditional RAM systems, whenever data is updated, the complete word (8-bit, 16-bit, or 32-bit) is rewritten even if only a single byte changes.
- This results in unnecessary power consumption, increased switching activity, and inefficient memory utilization.
- Traditional RAM designs are widely used in FPGA and ASIC applications because of their simple architecture and easy implementation using Verilog HDL.
- However, these systems lack selective byte-level control, making them less efficient for modern embedded systems, processors, cache memories, and communication applications where partial data modification is frequently required.

Block Diagram

1.Address Decoder

- Selects the required memory location using the input address.

2.Memory Array

- Stores binary data in memory cells.

3.Read/Write Control

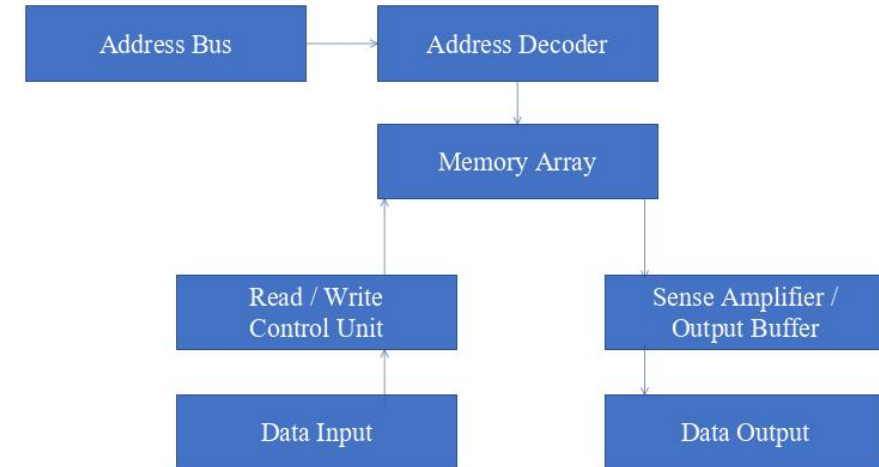
- Controls whether the RAM performs read or write operation.

4.Data Input

- Provides input data during write operation.

5.Data Output

- Produces stored data during read operation.



LIMITATION OF EXISTING SYSTEM

1. No support for byte-level access.
2. Performs full-word operations even for small data.
3. High power consumption due to unnecessary operations.
4. Increased memory access time.
5. Reduced overall system performance.
6. Not suitable for modern embedded systems.
7. Lacks flexibility and scalability.

PROPOSED SYSTEM

- The proposed system introduces a Byte-Enabled RAM Architecture designed using Verilog HDL for efficient and selective data access in FPGA and ASIC applications.
- Unlike the conventional RAM architecture, the proposed design allows individual bytes within a memory word to be independently read or written without modifying the remaining bytes.
- In traditional RAM systems, the complete memory word is rewritten even when only a small portion of data changes.
- This causes unnecessary switching activity, higher power consumption, and reduced efficiency. To overcome these limitations, the proposed architecture introduces Byte Enable Control Logic, which selectively activates only the required byte locations during write operations.

BLOCK DIAGRAM

1. Clock (clk)

- The clock signal synchronizes all operations in the RAM system.
- It controls when read and write actions occur.
- Data is stored or updated on the active clock edge.

2. Control Unit

- The control unit manages memory operations using control signals.

3. Byte Selection Logic

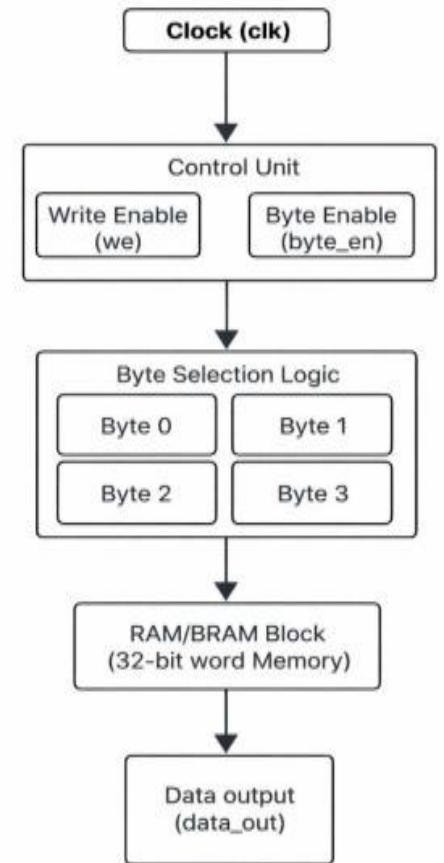
- Divides the 32-bit memory word into four 8-bit bytes. Only selected bytes are modified during write operation. Unselected bytes retain their previous data.

4. RAM/BRAM Block (32-bit Word Memory)

- Main memory storage unit of the architecture. Stores data in 32-bit word format. Reduces memory access overhead by enabling selective byte updates.

5. Data Output

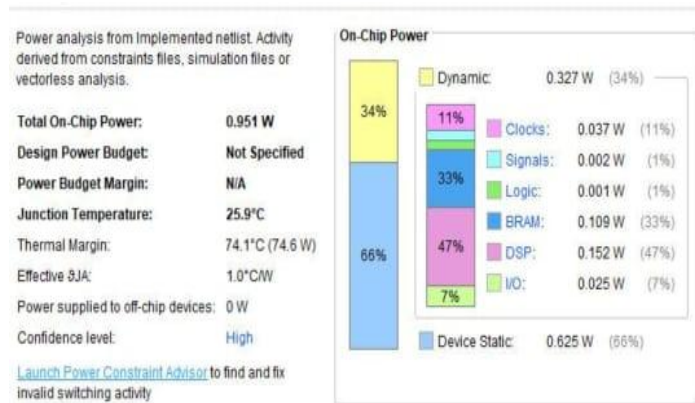
- Provides the stored data from memory.
- Outputs the 32-bit data word after read operation.
- Sends data to processor or external system.



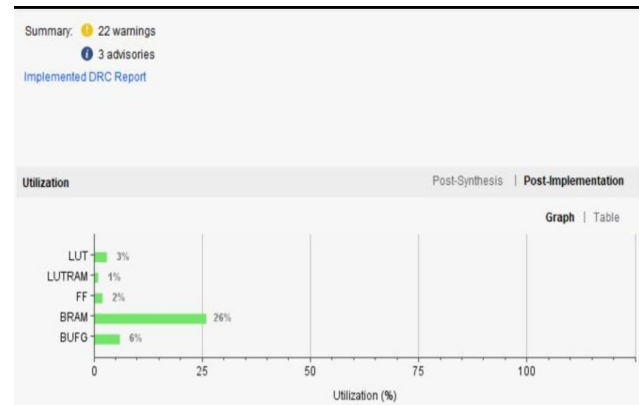
ADVANTAGE OF PROPOSED SYSTEM

- 1.Supports selective byte-level access.
2. Reduces unnecessary read and write operations.
3. Low power consumption.
4. Improves speed and overall performance.
5. Reduces switching activity inside memory.
6. Efficient use of memory bandwidth.
7. Flexible and scalable design.

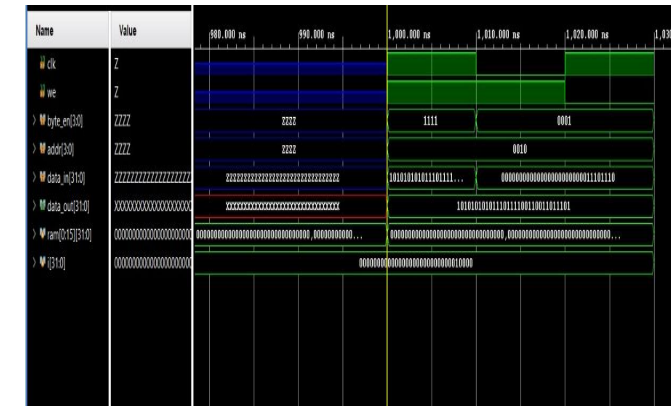
EXPERIMENTAL RESULT- EXISTING SYSTEM



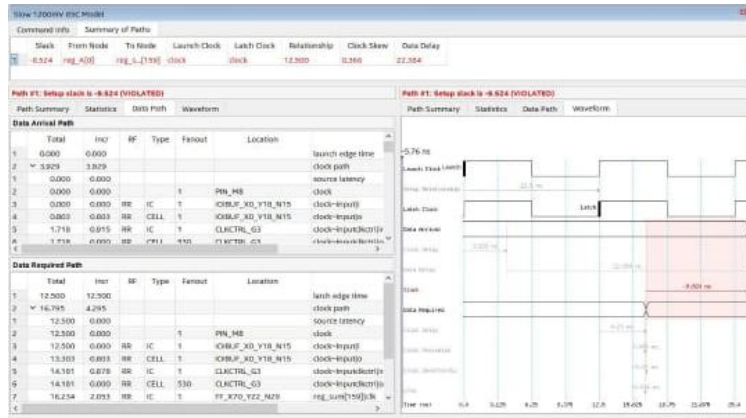
power analysis



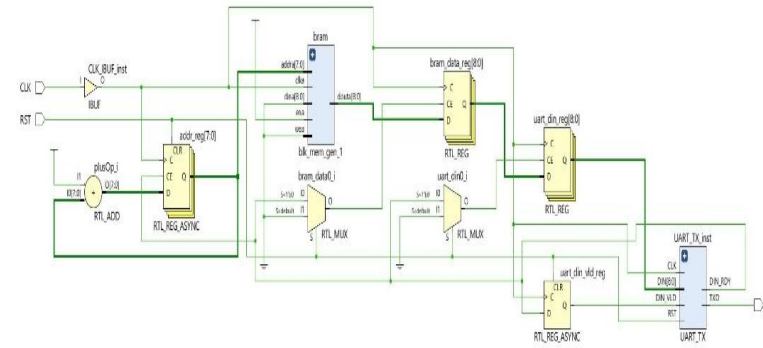
LUT report



simulation wave form

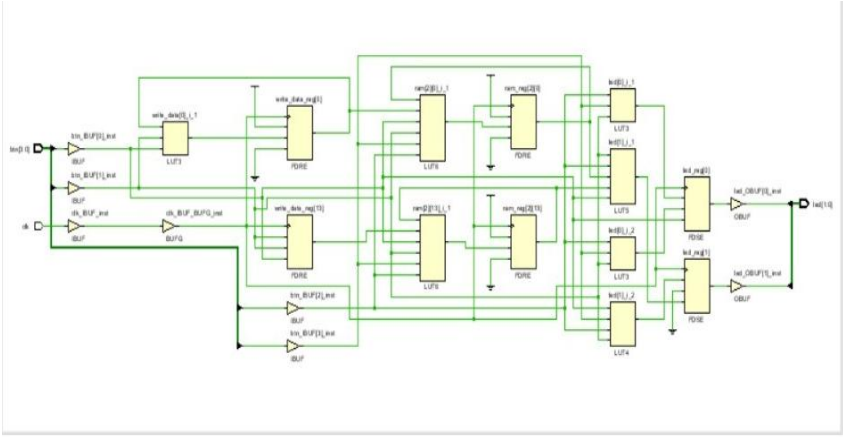


Delay

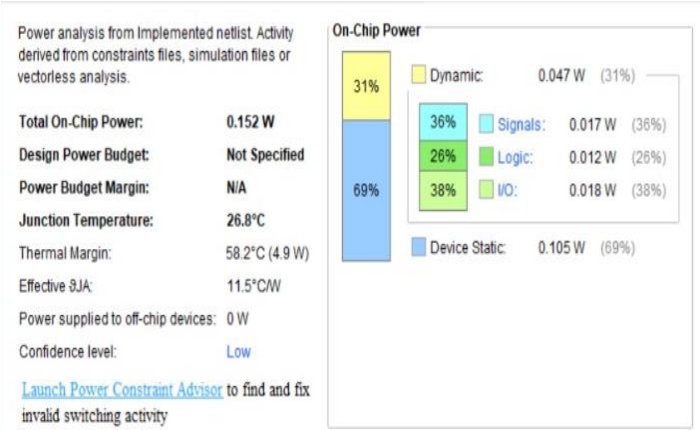


schematic diagram

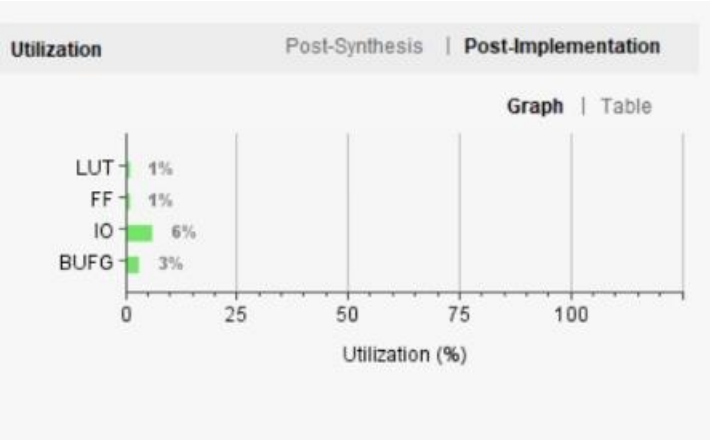
EXPERIMENTAL RESULT-PROPOSED SYSTEM



schematic diagram



power analysis



LUT report



simulation wave form

Name	Requirement	Path Delay	Logic Delay	Net Delay
Path 1	0	4.927	3.214	1.713
Path 2	0	4.885	3.213	1.672
Path 3	0	3.32	1.08	2.24
Path 4	0	3.15	1.08	2.07
Path 5	0	2.955	1.08	1.875
Path 6	0	2.841	1.116	1.726
Path 7	0	2.544	1.08	1.465
Path 8	0	2.544	1.08	1.465
Path 9	0	2.37	0.956	1.414
Path 10	0	2.221	0.956	1.266

Delay

COMPARISON OF EXISTING AND PROPOSED SYSTEMS

Parameter	Existing System	Proposed System
Memory Access	10 ns	6 ns
Data Transfer	Entire word transferred even for small data(69 %)	Only required byte is transferred (93 %)
Power Consumption	High due to unnecessary switching activity	Reduced power consumption
Read/Write Operation	Full word read/write only	Byte-wise read/write supported
Speed	80 MHz	128 MHz
Delay	22.94 pJ	8.73 pJ
Application Support	Suitable for basic memory systems	Suitable for ASIC, FPGA, cache, embedded systems

CONCLUSION

The byte-enabled RAM architecture was successfully designed using Verilog HDL to improve data access efficiency. It allows selective byte-level operations, reducing power consumption and unnecessary data processing. The system shows better performance, efficient memory usage, and high flexibility compared to conventional RAM. Simulation results confirm correct operation, making it suitable for , processors, and FPGA applications.

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